

ENG106-partb-writeup

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1 Brief Description

This is a spinoff of Andrej Karpathy's *nanogpt* using the Shakespeare corpus as training data.

I left model parameters **untouched** because I intended on fully training the model as I had enough time.

it took a bit over 3 hours to train the model for 2000 iterations.

2 Evaluation

Our model has a total of 10690625 parameters. We have a training loss of 1.139 and validation loss of 1.106. This is interesting to note as I'd expect validation loss to be higher.

We may be able to attribute this to randomness as well as Shakespeare's constant style may be more akin to an independent and identically distributed dataset.

When we generate text, we obtain:

ROMEO:

In this time of this grace of his high death.

ROMEO:

And let him from those and that high at him,
Could not seem against and hide his cheek,
Even to his allies brain-thoughts in himself.

Nurse:

Marry, I see the root happy hour;
For heaven shows hi

3 Takeaways

I was surprised by how simple the architecture was. I also watched a sizeable portion of Andrej Karparthy's 2 hour tutorial video.

If we switch the start text to math ("Compute two plus two"):

```
Compute two plus two another, that
says every face of our preyer; and, nor earth, I
see thee to take a more in marriage, let me be
seeing a top, and do to be with my fair. I
pardon, brain--a day shall not lie an hearing shows, but
therefore that feel a servant crownry t
```

As expected, we notice that the model has not learned logic in anyway. This makes me wonder how LLMs are trained on logic. Unlike Shakespeare, which is simply a style of connected words, logic is absolute and has far more degrees of freedom. I can assume that if we train on a large corpus of mathematical functions, it would not generalize well to an unseen function.